CASE STUDY 6 — CLOUD COMPUTING & ARCHITECTURE

Exploration of Cloud Service Models (IaaS, PaaS, SaaS)

DOMAIN	FREE-TIER PLATFORM	MODELS ANALYZED	KEY METRIC
Cloud Infrastructure	AWS / GCP / Azure	IaaS, PaaS, SaaS	Shared Responsibility

1. EXECUTIVE SUMMARY

Cloud computing provides scalable, on-demand compute and storage resources across three primary service models: **Infrastructure as a Service (IaaS)**, **Platform as a Service (PaaS)**, and **Software as a Service (SaaS)**. This case study evaluates the operational mechanics, resource governance, configuration overhead, and control distribution of each model using free-tier cloud environments (such as AWS Free Tier and Google Cloud Free Tier).

IaaS	PaaS	SaaS
FULL OS & INFRA CONTROL	MANAGED CODE DEPLOYMENT	FULLY MANAGED APPLICATION

2. CLOUD SHARED RESPONSIBILITY MODEL

The fundamental distinction among IaaS, PaaS, and SaaS lies in the division of management responsibilities between the cloud service provider (CSP) and the customer.

